

SNEHA SONKUSARE

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SUMMARY

Data Scientist with 3+ years of experience in ML, deep learning, NLP, and Generative AI across financial services and research. Built and deployed end-to-end pipelines ranging from XGBoost models to LLM fine-tuning (LoRA/QLoRA) and RAG systems. Proficient in Python, SQL, PyTorch, TensorFlow, AWS. M.S. in Data Science.

SKILLS

- Languages:** Python (Pandas, NumPy, Scikit-learn), R, SQL
- Machine Learning:** Classification, Regression, Clustering, Random Forest, Feature Engineering, XGBoost, LightGBM, SVMs
- Deep Learning:** TensorFlow, Keras, PyTorch, CNNs, LSTMs, NLP models
- Generative AI:** LLMs, RAG Pipeline, LLaMa 3.3 70B, DeepSeek-R1, Groq API, LangChain, LoRA/QLoRA, Prompt Engineering, Neo4j
- Visualization:** Tableau, Power BI, Seaborn, Matplotlib
- Databases:** MySQL, Snowflake, Amazon Redshift, PostgreSQL
- Cloud & Tools:** AWS (S3, Glue, SageMaker, QuickSight), Apache Spark, Airflow, dbt, Docker, CI/CD, Streamlit
- Statistics:** Hypothesis Testing, A/B Testing, Regression Analysis, Bayesian Inference, Price Elasticity Modeling

EXPERIENCE

Data Scientist (Research Assistant) | UMass School of Marine Science & Technology New Bedford, MA | Jan 2025 – Dec 2025

- Performed EDA on 500K+ oceanographic records, engineered domain-specific features, and trained Random Forest and CNN models (PyTorch/TensorFlow) achieving 85% accuracy; tracked all experiments via MLflow for full reproducibility.
- Optimized model performance through cross-validation, precision, recall, and F1 tuning; translated findings into actionable insights for non-technical research teams, directly informing conservation decisions.
- Spearheaded standardization of metadata and data governance practices across 5+ research systems that lacked consistent structure cutting data inconsistencies by 25% and ensuring model reproducibility across research and IT teams.
- Developed Apache Spark pipelines for end-to-end sensor data processing, collaborating with marine scientists to align model outputs with research needs and improving data accuracy by 20%.

Data Analyst | Hexaware Technologies LTD Mumbai, India | Sep 2021 – Jan 2024

- Collaborated with finance, operations, and product stakeholders to translate business pain points into data questions, writing complex SQL queries (joins, aggregations, window functions) across 5+ source systems to build a unified data layer, reducing data errors by 30% and saving 10+ hours of weekly manual corrections.
- Detected recurring anomalies in fraud and credit risk data through EDA and statistical hypothesis testing; applied Scikit-learn classification models that improved decision accuracy by 25%, and maintained 8+ Power BI dashboards for fraud detection, risk, and portfolio monitoring, cutting report generation time by 35%.
- Established 15+ KPIs across finance, product, and operations in collaboration with senior stakeholders, delivering self-serve analytics that reduced ad-hoc data requests by 20% and eliminated recurring manual reporting rebuilds for analysts.
- Diagnosed and resolved 20+ pipeline failures in Agile cross-functional teams alongside senior engineers, applying targeted fixes that boosted processing speed by 30% and reduced query runtime by 40%, while clearly communicating issue impact to non-technical stakeholders throughout.

PROJECTS

AI-Powered Learning Platform | [LearnAI](#) Feb 2026 – Mar 2026

- Developed and deployed an AI learning platform using LLaMA 3.3 70B and Groq API, generating personalized courses from a single input with 10+ tools including Mock Interview, Quiz Generator, and AI Doubt Solver.
- Engineered a structured prompt system that consolidated 7 API calls into 1, reducing generation time by 80%, and implemented a 3-model fallback chain for rate-limit resilience with session-state caching across 12-chapter multilingual lessons.

Job Market Intelligence and Career Advisor | [JobMarketIntel](#) Oct 2025 – Dec 2025

- Engineered a real-time pipeline that scrapes hundreds of daily job postings, extracts in-demand skills using NLP, and predicts salaries using XGBoost with SHAP explainability.
- Built a Personal Career Advisor that inputs a user's skillset, predicts their market rate, and recommends the top 3 skills to maximize salary growth powered by a custom Job Market Health Index (0–100 scale).

Instruction Fine-Tuning Based Medical Chatbot | [GitHub](#) Apr 2025 – May 2025

- Fine-tuned DeepSeek-R1 (LoRA/QLoRA) on medical Q&A and built a Neo4j-backed RAG pipeline for structured clinical data retrieval, improving response accuracy by 30%.
- Developed a production-ready inference pipeline using Flask REST API with real-time model monitoring for latency and drift, containerized and configured for AWS deployment.

EDUCATION

UNIVERSITY OF MASSACHUSETTS DARTMOUTH Jan 2024 – Dec 2025

Master of Science in Data Science | GPA : 3.73

(Relevant Coursework: Artificial Intelligence, Advanced Machine Learning, Big Data Analytics, Database Design).

MAHARASHTRA INSTITUTE OF TECHNOLOGY

Aug 2017 – Jul 2021

Bachelor of Technology – Electronics and Communication